PROJECT WORK ASSIGNMENT

FitBuddy – AI Fitness Plan Generator using Gemini Models

Team ID	unm293200909
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Team Member	Kavyasri.A

Project Work Assigned to Kavyasri.A

Since Kavyasri.A is the only team member provided, the complete FitBuddy project workflow is assigned to Kavyasri.A. The responsibilities below are derived from the project document and cover model selection, architecture, development, routing, frontend, database integration, testing, deployment, and final verification.

Work Order	Assigned Member	Assigned Work
1	Kavyasri.A	**Model Selection & Architecture:** Research and select Gemini models; define the FastAPI-based architecture; set up the Python/virtual environment and required libraries.
2	Kavyasri.A	**Core AI Functionality:** Implement personalized 7-day workout generation using Gemini 1.5 Pro, nutrition/recovery tips using Gemini Flash, and feedback-based plan updating.
3	Kavyasri.A	**Backend & Routing:** Develop routes.py, process form input, validate data with Pydantic, connect Gemini functions, save user/plan data, and implement the admin view.
4	Kavyasri.A	**Database:** Implement SQLite/SQLAlchemy persistence for user information, original plans, and feedback-based updated plans.
5	Kavyasri.A	**Frontend Development:** Build index.html, result.html and all_users.html; create responsive CSS layouts and integrate dynamic Jinja2 templates.
6	Kavyasri.A	**Deployment & Testing:** Configure environment variables, install requirements, run the FastAPI/Uvicorn server, test the web interface and /docs API documentation.
7	Kavyasri.A	**Final Verification:** Verify workout generation, nutrition tips, feedback updates, data storage, admin display, and overall user flow; prepare the project for demonstration.

Work Allocation by Project Milestones

Milestone	Activities Assigned to Kavyasri.A
Milestone 1 – Model Selection & Architecture	Research/select Gemini models; define application architecture; set up development environment.
Milestone 2 – Core Functionalities	Develop workout generation, nutrition/recovery tips, feedback-based plan updating, user/plan storage, and FastAPI input processing.
Milestone 3 – routes.py Development	Implement home, workout generation, feedback update, and admin routes; integrate Gemini responses and database functions.
Milestone 4 – Frontend Development	Design responsive user interface and create dynamic Jinja2 templates for user input, results, feedback, and admin view.
Milestone 5 – Deployment	Prepare local deployment, configure API key environment variables, run Uvicorn, and test the application and API documentation.

Expected Deliverables

• Functional Gemini-powered 7-day personalized workout planner.

• Goal-specific nutrition or recovery tip generation.

- Feedback-based workout plan revision.
- FastAPI backend with routes and validated user input.
- SQLite database with SQLAlchemy for user and plan storage.
- Responsive HTML/CSS/Jinja2 frontend and admin dashboard.
- Local deployment and testing through Uvicorn and FastAPI /docs.

Reference basis: The FitBuddy project document describes the Gemini Pro/Flash integration, FastAPI backend, SQLite/SQLAlchemy storage, Jinja2 frontend, project milestones, and deployment/testing activities.